

Tanushree Burman

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RESEARCH OVERVIEW

My research focuses on human-in-the-loop reinforcement learning, investigating how demonstrations and trajectory preferences can be used to improve the performance and alignment of reinforcement learning agents. In particular, I study how user preferences and interaction costs affect feedback quality and agent learning. My prior work includes developing interactive robotics systems for Machine Learning and Reinforcement Learning education.

EDUCATION

Tufts University, Medford, MA

Expected Graduation: May 2028

Ph.D. Candidate in Computer Science

Advisor: Jivko Sinapov

Tufts University, Medford, MA

August 2022 – May 2025

Master of Science in Computer Science; GPA: 3.7

Relevant Coursework: Reinforcement Learning, Human-Robot Interaction, Robot Operating System (ROS), Machine Learning, Probabilistic Robotics

Yale University, New Haven, CT

August 2018 – May 2022

Bachelor of Science in Electrical Engineering and Computer Science; GPA: 3.5

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, NumPy, Pandas), C, Java, SQL, JavaScript, HTML, CSS

Robotic Frameworks and Simulators: Robot Operating System (ROS), Kinova Gen 3 Lite, OpenAI Gym, PyBullet, Unity, RViz, MoveIt

Software & Tools: AI Prompt Engineering (Claude, GPT), Git, Unix

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, OpenCV

INDUSTRY EXPERIENCE

Lockheed Martin, Shelton, CT

June 2022 – August 2022

AI Machine Learning Engineer Intern

- Customized a PyBullet-based quadrotor simulation environment for the MIMUA (Multimodal Interactive Manned-Unmanned Agent) project by integrating URDF and OBJ/DAE 3D models and developing realistic military terrain, textures, and obstacles for multimodal reinforcement learning.
- Implemented coordinate-space transformations to expand the effective training state space, improving agent generalization while reducing the computational cost of training across the full coordinate space.

AI Machine Learning Engineer Intern

June 2021 – December 2021

- Investigated Population-Based Training (PBT) for deep reinforcement learning, using player attributes to parameterize game environments and train diverse agent populations to study game balance and agent performance.
- Built interactive Streamlit and Bokeh dashboards to visualize agent performance and analyze training outcomes across RL experiments.

RESEARCH EXPERIENCE

Tufts University, Medford, MA

August 2022 – Present

Research Assistant

- Conduct research in the Multimodal Learning, Interaction and Perception Lab: Investigating how different modalities of human feedback—demonstrations and preferences over trajectories—can be incorporated into reinforcement learning to improve agent performance, with a central focus on how user experience (perceived ease and effort) shape feedback quality and should inform the design of preference-based learning systems.

Tufts Center for Engineering Education and Outreach, Medford, MA

September 2022 – August 2024

Research Assistant

- Contributed to the development of Smart Motor, an educational robotics toolkit introducing AI concepts such as supervised machine learning and reinforcement learning to elementary and high school students. The toolkit was used in elementary school robotics summer programs serving more than 200 students.

PUBLICATIONS

Peer-Reviewed Conference Papers

- [C-3] **Burman, T.**, Dahal, M., Xu, G., Rogers, C., Cross, J., & Sinapov, J. (2025). Smart Motor: A Low-Cost Hardware and Software Toolkit for Introducing Supervised Machine Learning to Elementary School Students. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(28), 29128–29136. [Link to Paper](#).
- [C-2] **Burman, T.**, Coney, J., Rogers, C., Cross, J., Sinapov, J. (2025). Balancing Act: Mastering Beam-and-Ball Control with Reinforcement Learning. In: Balogh, R., Obdržálek, D., Fachantidis, N. (eds) *Robotics in Education*. RiE 2025. Lecture Notes in Networks and Systems, vol 1544. Springer, Cham. [Link to Paper](#).
- [C-1] Shukla, Y., **Burman, T.**, Kulkarni, A. N., Wright, R., Velasquez, A., & Sinapov, J. (2024). Logical Specifications-guided Dynamic Task Sampling for Reinforcement Learning Agents. *Proceedings of the International Conference on Automated Planning and Scheduling*, 34(1), 532–540. [Link to Paper](#).

Peer-Reviewed Journal Articles

- [J-1] Huang, J., Shung, D., **Burman, T.**, Krishnaswamy, S., & Batra, R. K. (2022). Prediction of Death after Terminal Extubation, the Machine Learning Way. *Journal of the American College of Surgeons*, 235(5), S95–S96. [Link to Paper](#).

PERSONAL AWARDS

- [2] **Graduate Student Council Service Award**, Tufts University 2026
This award recognizes a graduate student for outstanding contributions to graduate student life through academic and professional programming, community outreach and service, and social events.
- [1] **Robert P. Guertin Graduate Student Leadership Award Finalist**, Tufts University 2026
This award recognizes outstanding graduate student leadership across the arts, humanities, sciences, and engineering — including contributions like academic/social programming, curricular initiatives and peer support.

TEACHING EXPERIENCE

CS 135 Machine Learning and Data Mining, Tufts University September 2024 – Present
Teaching Assistant

- Taught and supported 350+ students across 7 semesters through weekly office hours, debugging, and project guidance in software development and machine learning, including regression, classification, neural networks, and recommendation systems; evaluated and graded student projects.

Pre-College High School Summer Program, Tufts University July 2024 – July 2026
Instructor

- Designed and taught a reinforcement learning curriculum across 5 sessions for 150+ high school students, developing coding and hands-on robotics projects to introduce core RL concepts through interactive learning.

Girls Who Code June 2023
Instructor

- Led a virtual web development course for 60 high school students, supervising 5 teaching assistants and providing technical instruction, hands-on programming activities, and debugging support.

Teaching Assistant July 2020 – August 2020

- Delivered virtual lessons and office hours for 30 high school students, teaching HTML, CSS, and JavaScript and providing technical and debugging support for programming assignments.

UNIVERSITY SERVICE & LEADERSHIP

Tufts Graduate Student Council April 2025 – May 2026
President

- Led the Tufts Graduate Student Council, expanding graduate orientation initiatives and developing institutional partnerships to increase student engagement and connection across the Tufts community.

Student Life Chair April 2023 – April 2025

- Coordinated and led community-building events for 1,500+ graduate students, strengthening engagement and connection across the graduate student community.